

VIT-AP
UNIVERSITY

DIGITAL LOGIC DESIGN
(Course Code: ECE 1003)

Module-2:Lecture-5

Combinational Circuit Design

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Module-2

- ❖ Binary adder and subtractor
- ❖ Ripple carry adders/subtractors and fast adders
- ❖ Binary decoders, encoders
- ❖ **Multiplexers and de-multiplexers**
- ❖ **Logic functions using** decoders and **multiplexers**
- ❖ Code converters
- ❖ Magnitude comparator

CONTENTS

Lecture-5

- ❖ Higher order Multiplexer using Lower order Multiplexer
- ❖ Boolean function Implementation using Multiplexer
- ❖ Demultiplexer

BOOKS

Textbooks

1. M.Morris Mano, Michael D Ciletti, **Digital Design**, 5th edition, Pearson Publishers, 2013.
2. R.P. Jain, “**Modern Digital Electronics**”, 4th edition, TMH.

References

1. M.Morris Mano, Charles R. Kime, Tom Martin, **Logic and Computer Design Fundamentals**, 4th edition, Pearson Publishers.
2. C. H. Roth and L. L. Kinney, **Fundamentals of Logic Design**, 5th edition, Cengage Publishers.

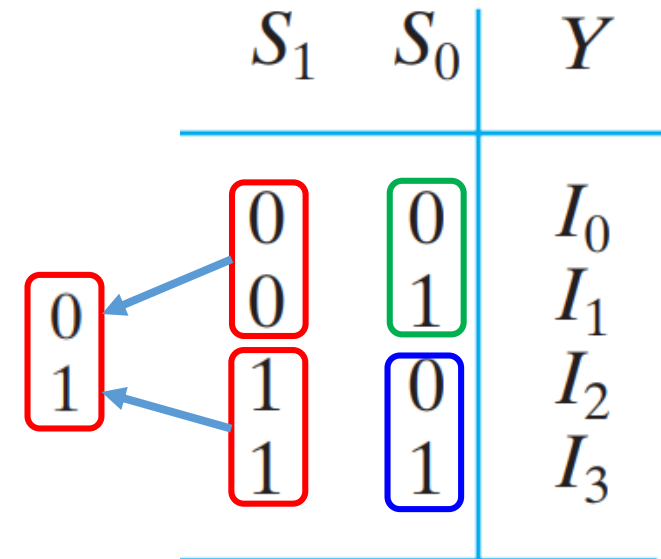
Multiplexer

MULTIPLEXER

4x1 Multiplexer using 2x1 Multiplexer

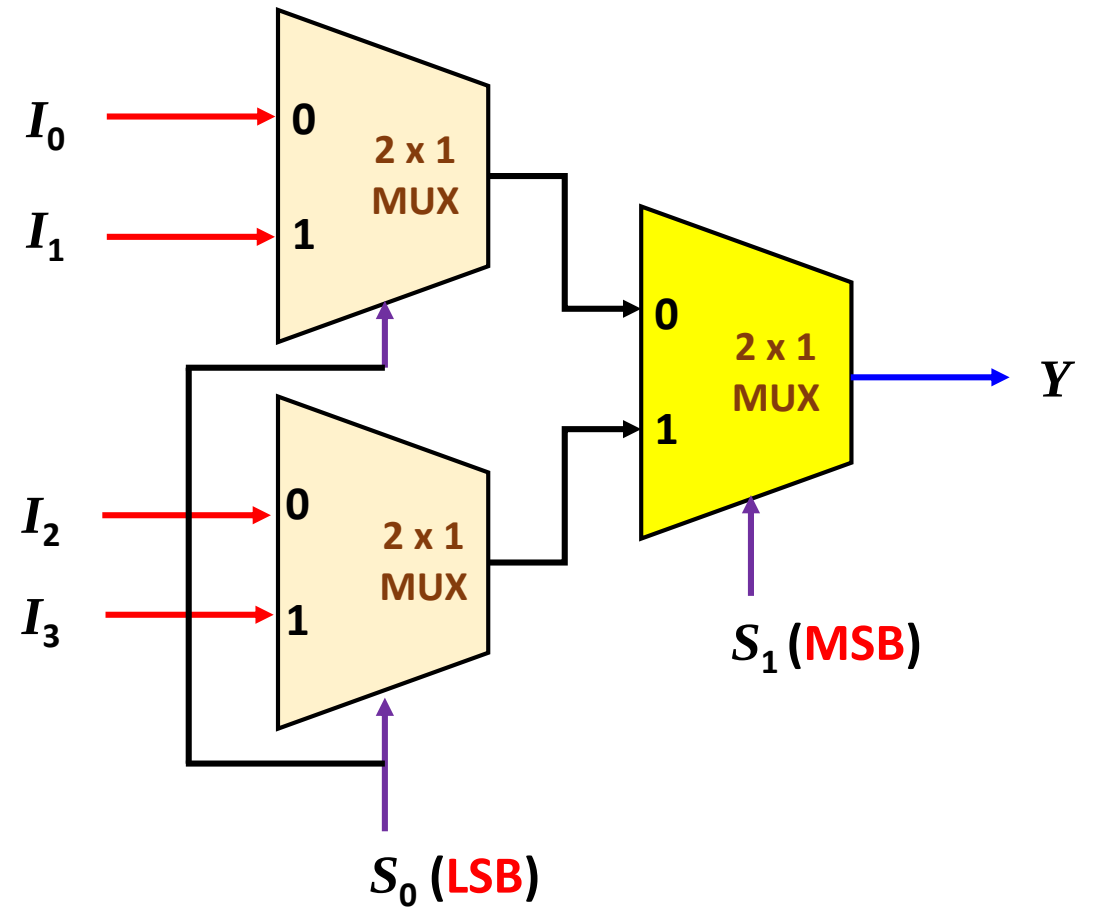
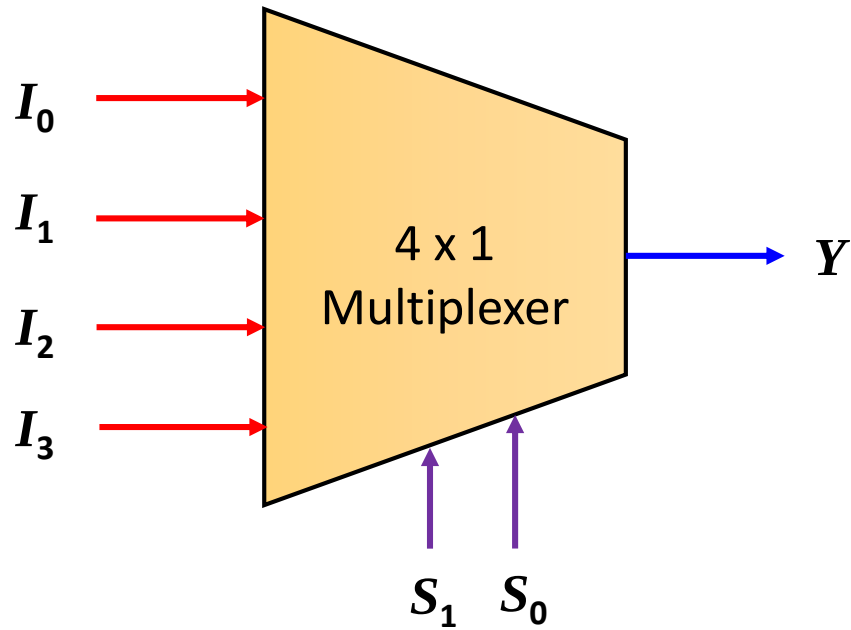
- ✓ To design a **4x1 multiplexer**, we require **three 2x1 multiplexers**.
- ✓ Multiplexer 1 is to choose I_0 and I_1
- ✓ Multiplexer 2 is to choose I_2 and I_3
- ✓ Multiplexer 3 is to choose multiplexer 1 and multiplexer 2
- ✓ There are two selection pins S_1 and S_0

S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3



MULTIPLEXER

4x1 Multiplexer using 2x1 Multiplexer

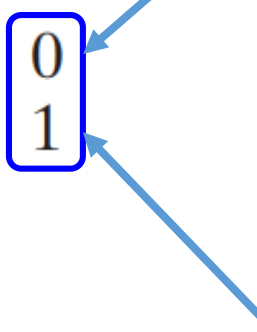


MULTIPLEXER

8x1 Multiplexer using both 4x1 & 2x1 Multiplexer

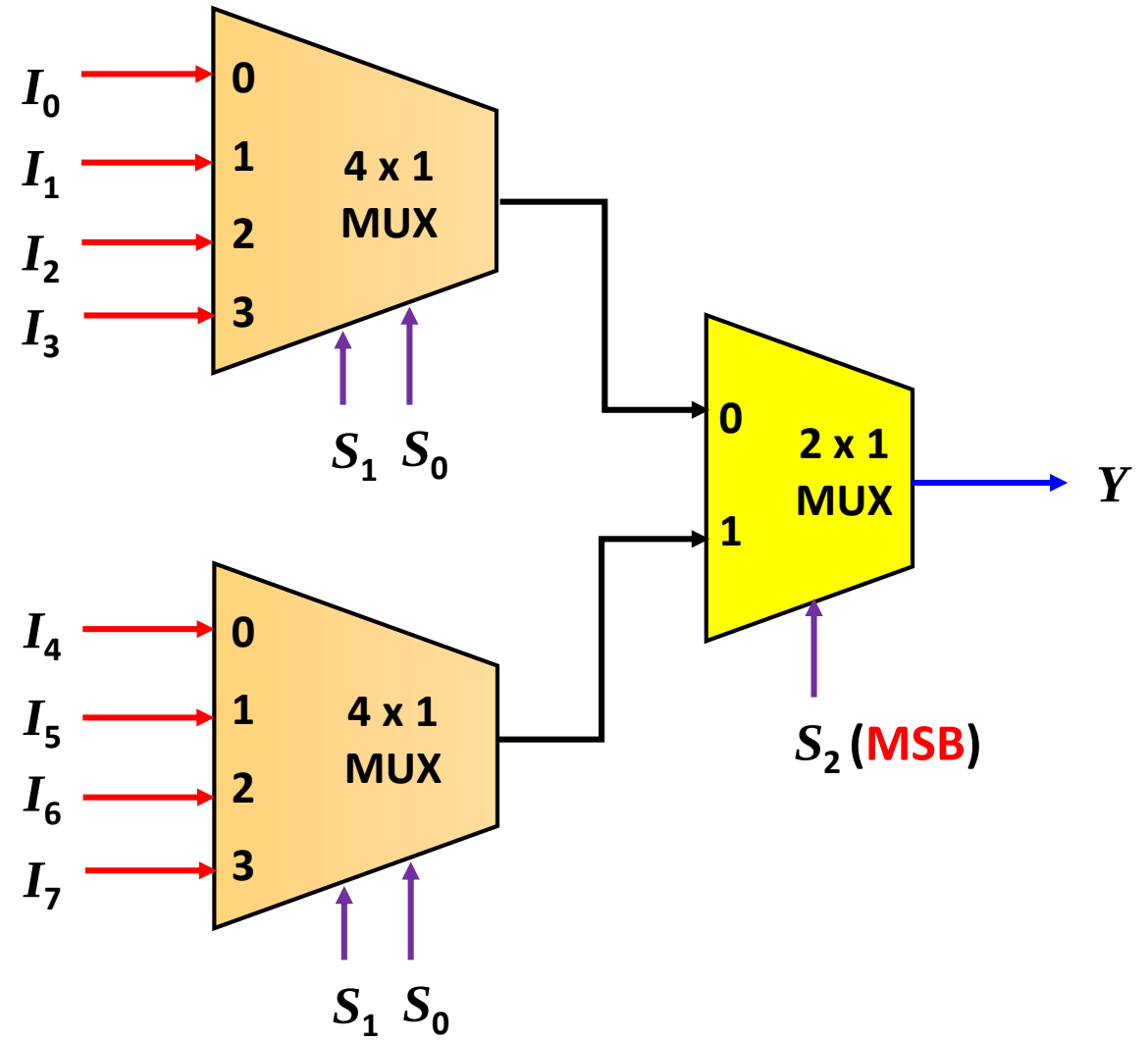
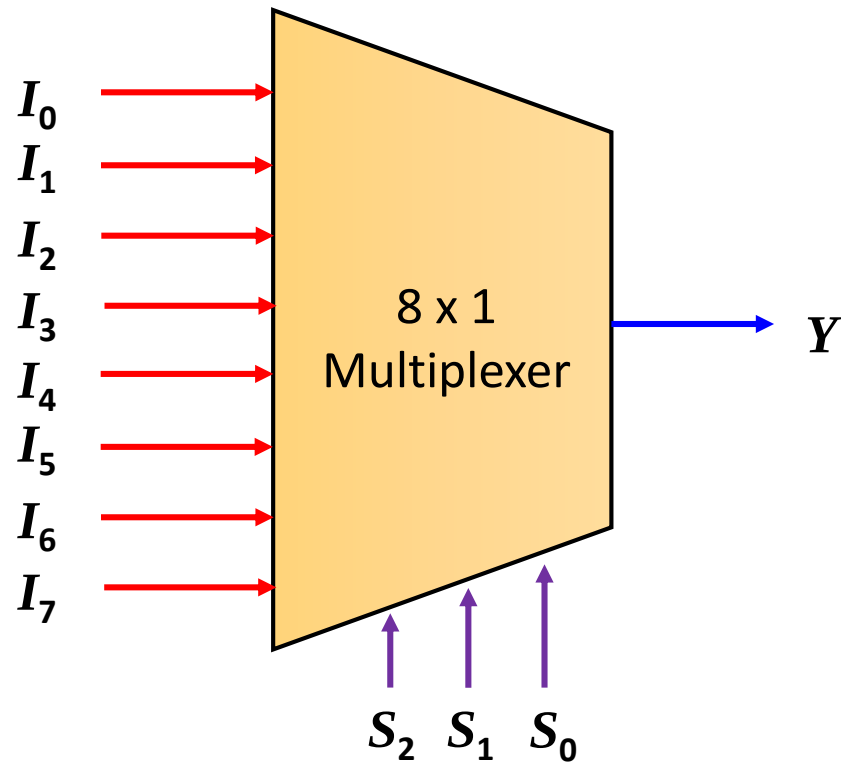
- ✓ To design an **8x1 multiplexer**, we require **two 4x1 multiplexers** and one 2x1 multiplexer is used
- ✓ 4x1 Multiplexer 1 is to choose I_0 to I_3
- ✓ 4x1 Multiplexer 2 is to choose I_4 to I_7
- ✓ 2x1 Multiplexer is to choose 4x1 multiplexer 1 and 4x1 multiplexer 2
- ✓ There are three selection pins S_2 , S_1 and S_0

S_2	S_1	S_0	Y
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7



MULTIPLEXER

8x1 Multiplexer using both 4x1 & 2x1 Multiplexer



MULTIPLEXER

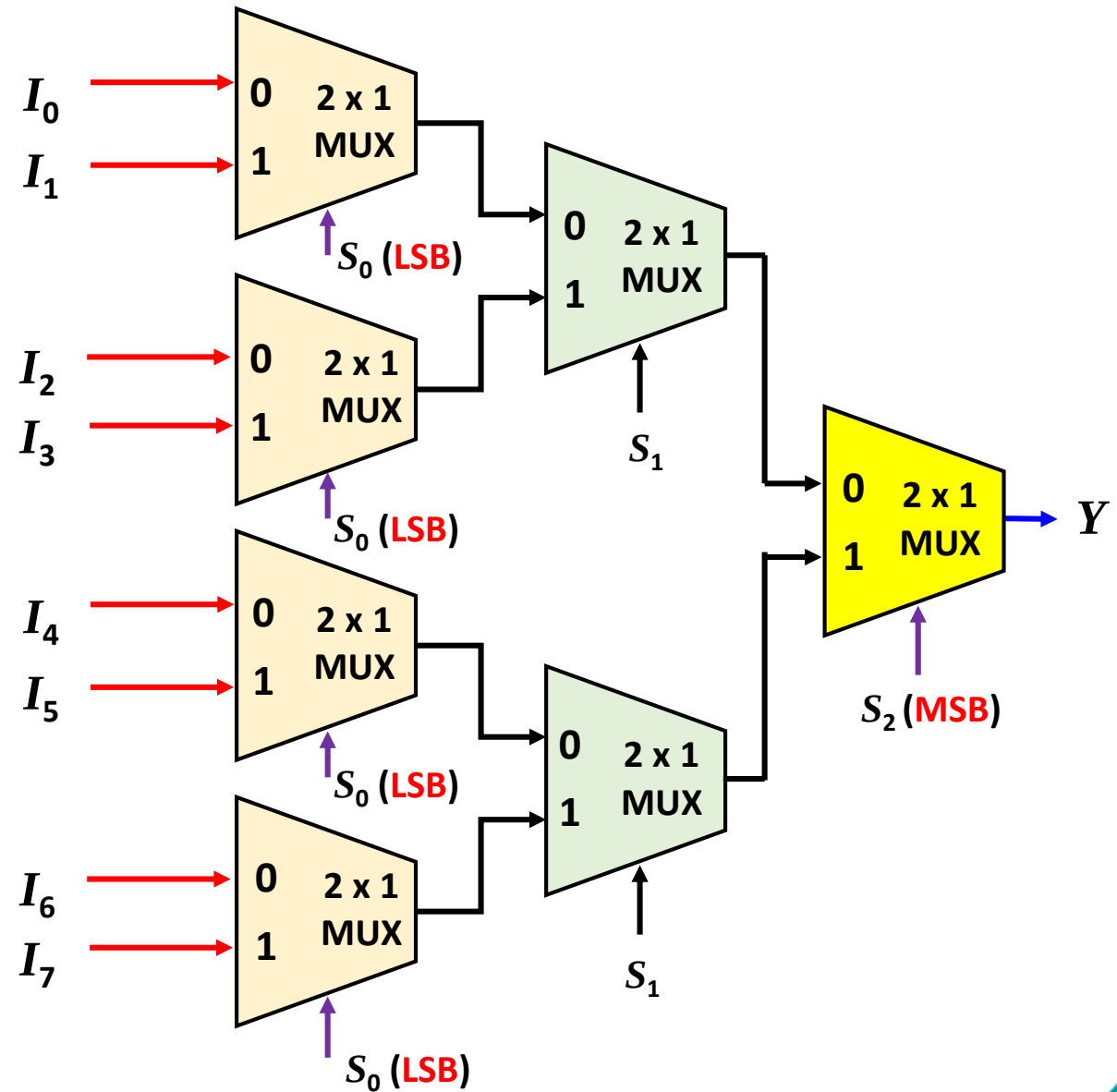
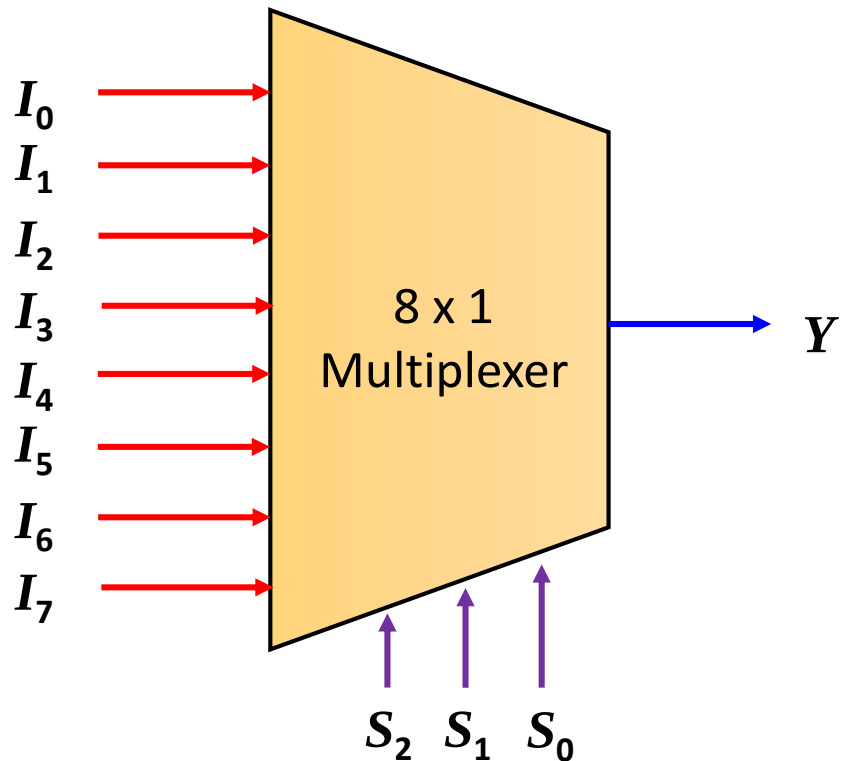
8x1 Multiplexer using 2x1 Multiplexer

- ✓ To design an **8x1 multiplexer**, we require seven 2x1 multiplexer is used
- ✓ 2x1 Multiplexer 1 is to choose I_0 to I_1
- ✓ 2x1 Multiplexer 2 is to choose I_2 to I_3
- ✓ 2x1 Multiplexer 3 is to choose I_4 to I_5
- ✓ 2x1 Multiplexer 4 is to choose I_6 to I_7
- ✓ 2x1 Multiplexer 5 is to choose 2x1 multiplexer 1 and 2x1 multiplexer 2
- ✓ 2x1 Multiplexer 6 is to choose 2x1 multiplexer 3 and 2x1 multiplexer 4
- ✓ 2x1 Multiplexer 7 is to choose 2x1 multiplexer 5 and 2x1 multiplexer 6
- ✓ There are three selection pins S_2 , S_1 and S_0

S_2	S_1	S_0	Y
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7

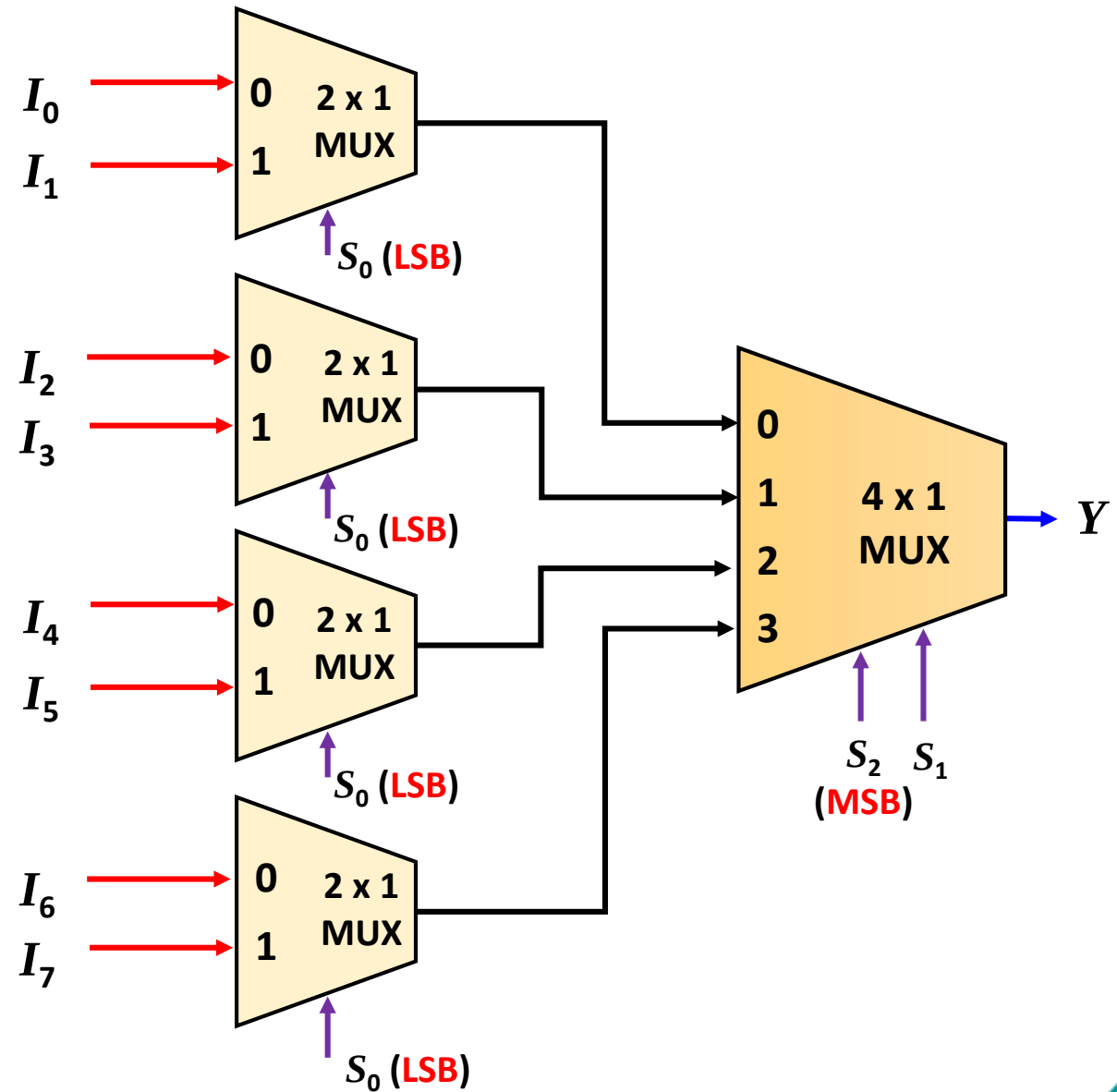
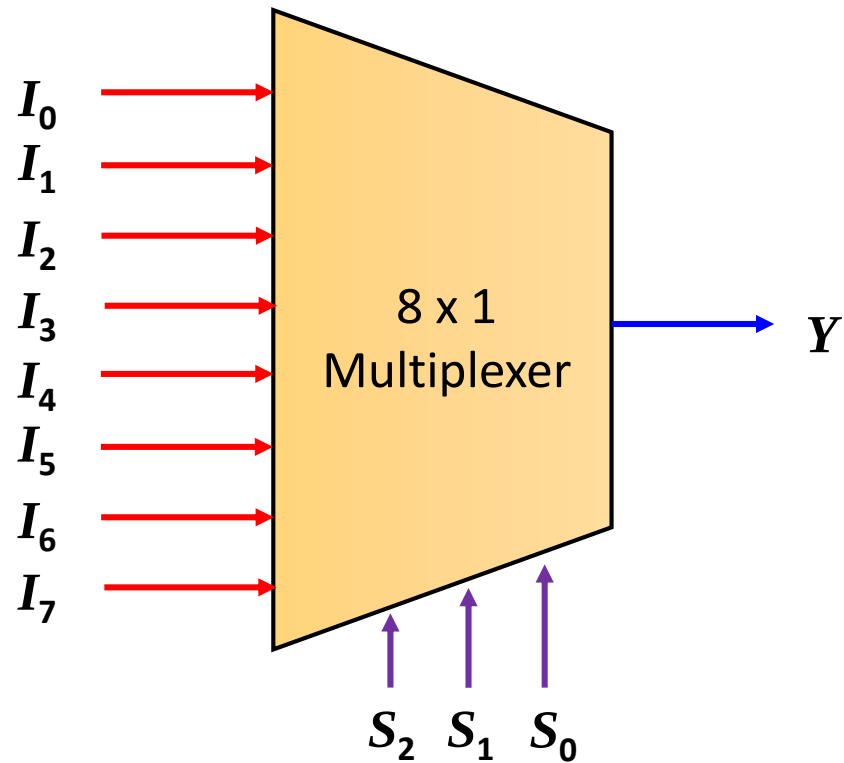
MULTIPLEXER

8x1 Multiplexer using 2x1 Multiplexer



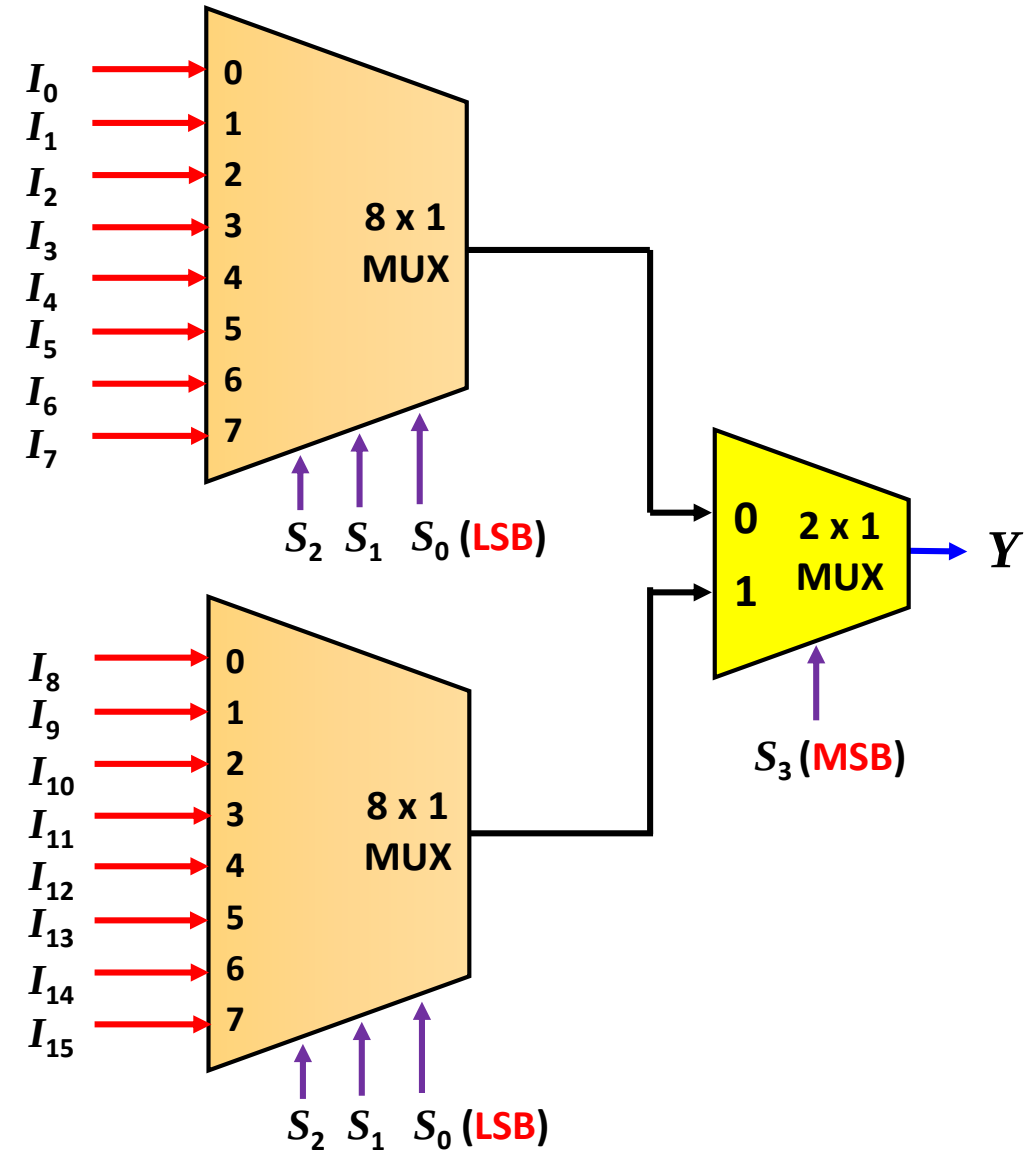
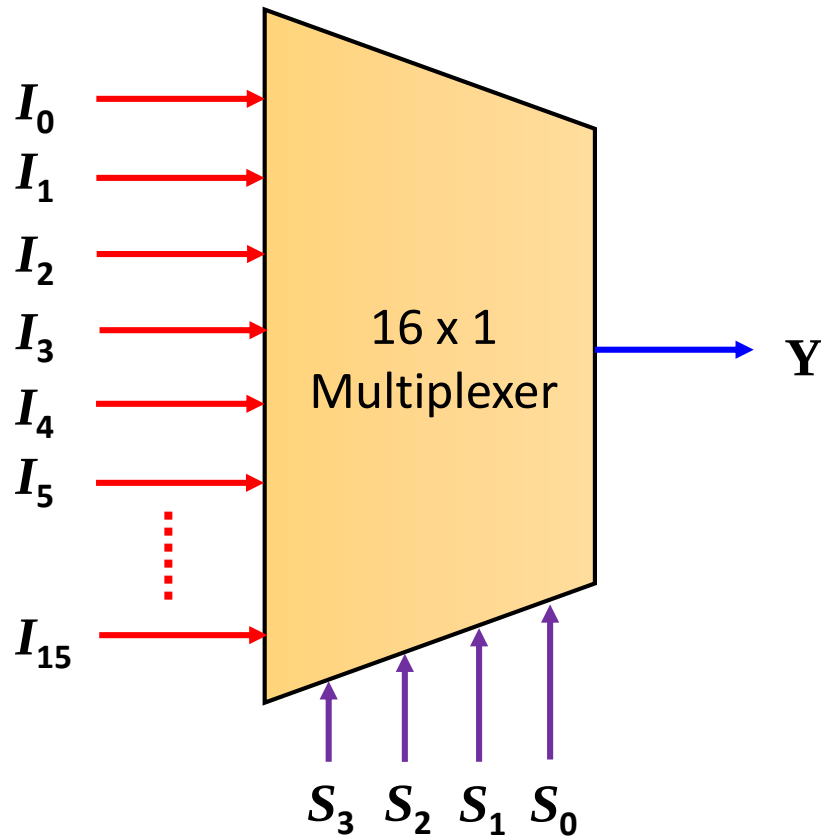
MULTIPLEXER

8x1 Multiplexer using both 2x1 & 4x1 Multiplexer



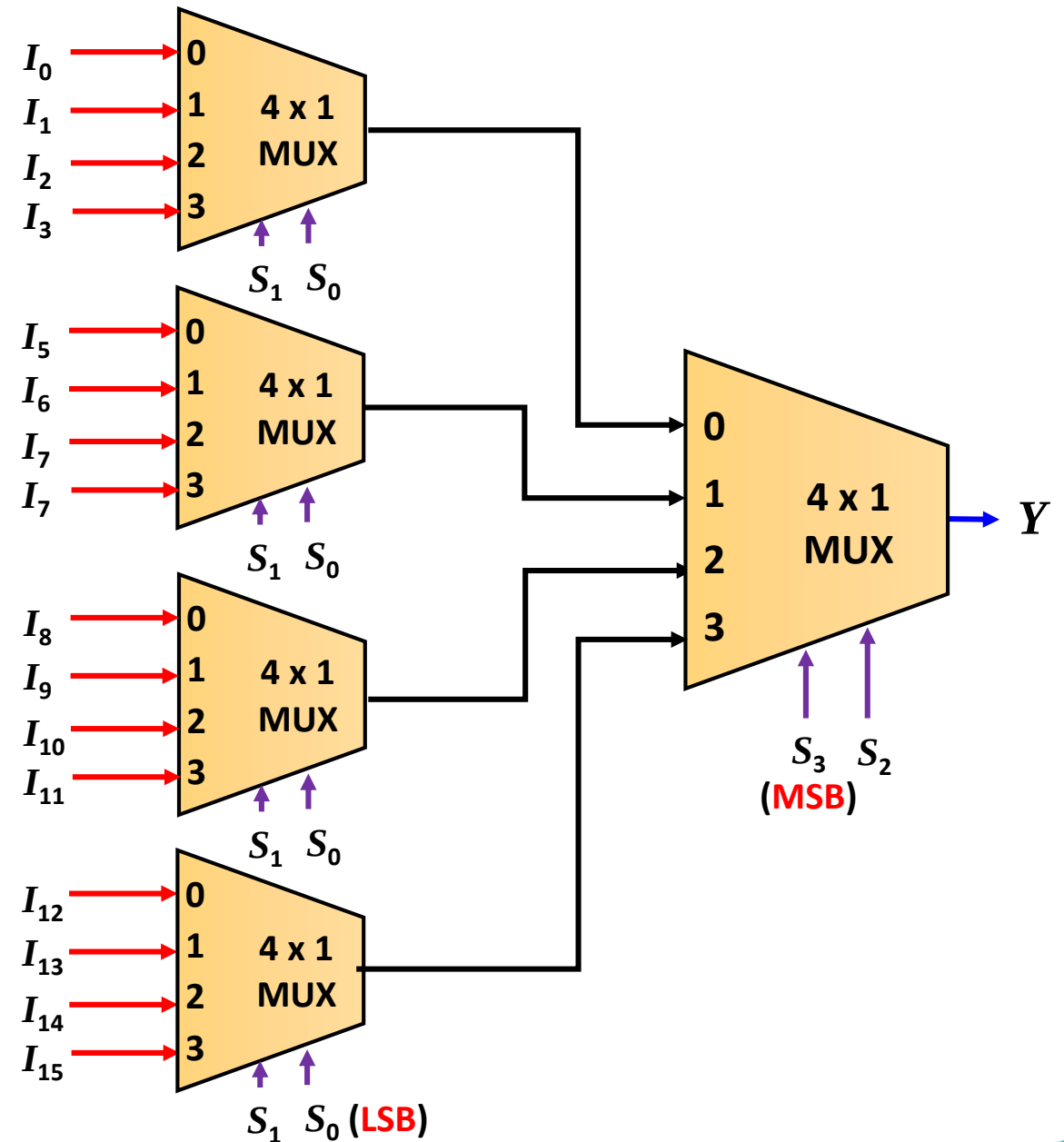
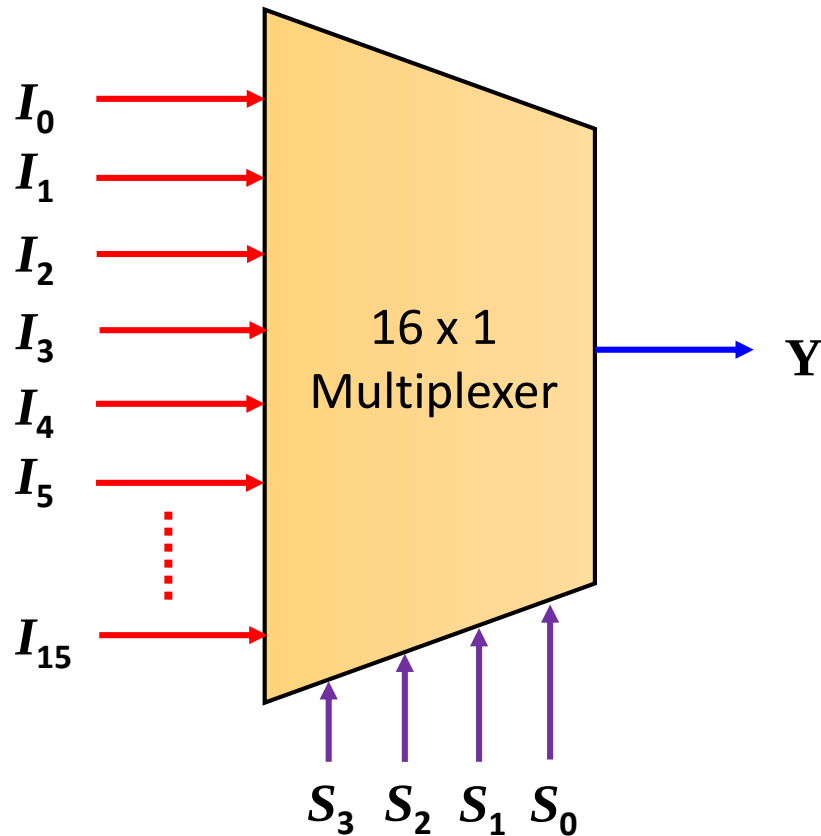
MULTIPLEXER

16x1 Multiplexer using 8x1 and 2x1 Multiplexer



MULTIPLEXER

16x1 Multiplexer using 4x1 Multiplexer



MULTIPLEXER

Question: Design the following Boolean function using appropriate Multiplexer and logic gates

$$F(x, y, z) = \Sigma(1, 2, 6, 7)$$

Solution:

Step-1: Number of variables (n) = 3

Then, the number of selection input = $n - 1 = 3 - 1 = 2$

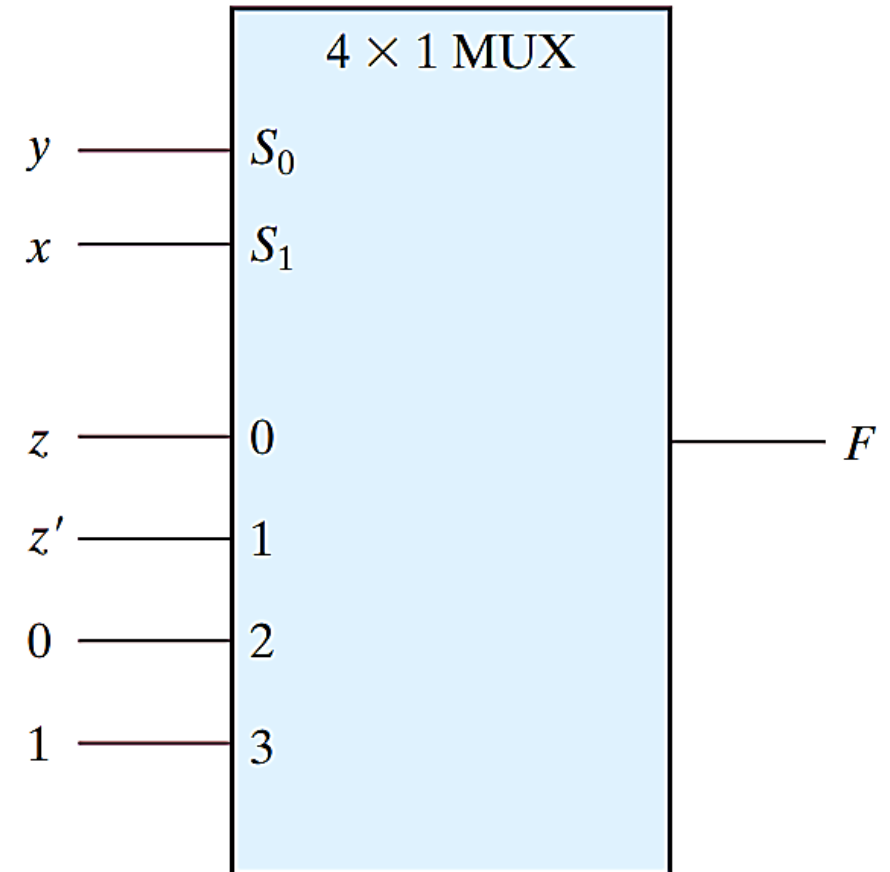
Step-2: Choose x and y as selection input s_1 and s_0
output F as function of third variable z

MULTIPLEXER

Step-3: Truth table

		s_1	s_0				
		x	y	z	F		
s_1	s_0	{	0	0	0	0	$F = z$
0	0		0	0	1	1	
0	1	{	0	1	0	1	$F = z'$
			0	1	1	0	
1	0	{	1	0	0	0	$F = 0$
			1	0	1	0	
1	1	{	1	1	0	1	$F = 1$
			1	1	1	1	

Step-4: Circuit Design



MULTIPLEXER

Question: Design the following Boolean function using appropriate Multiplexer and logic gates

$$F(A, B, C, D) = \Sigma(1, 3, 4, 11, 12, 13, 14, 15)$$

Solution:

Step-1: Number of variables (n) = 4

Then, the number of selection input = $n - 1 = 4 - 1 = 3$

Step-2: Choose *A*, *B* and *C* as selection input s_2 , s_1 and s_0
output *F* as function of forth variable *D*

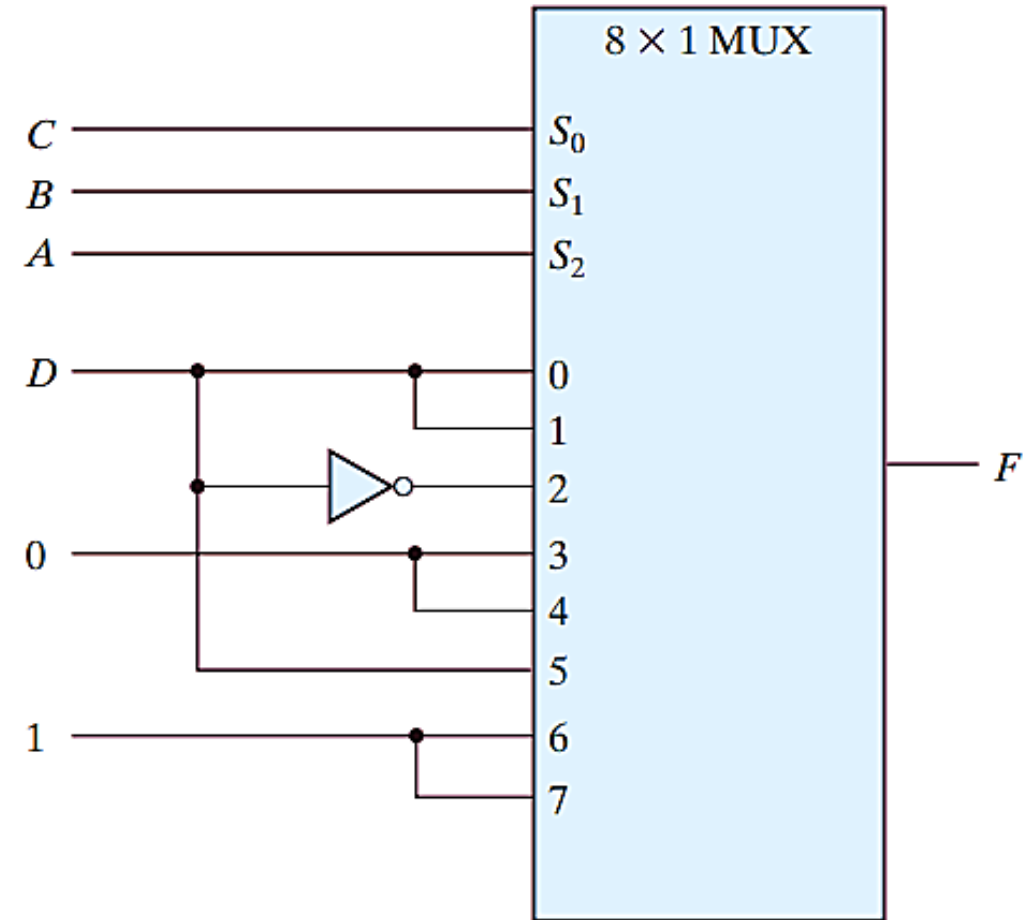
MULTIPLEXER

Step-3:

Truth table

s_2	s_1	s_0	s_2	s_1	s_0	A	B	C	D	F	
0	0	0	{	0	0	0	0	0	0	0	$F = D$
				0	0	0	0	1	1	1	
0	0	1	{	0	0	1	0	0	0	0	$F = D$
				0	0	1	0	1	1	1	
0	1	0	{	0	1	0	0	0	0	1	$F = D'$
				0	1	0	0	1	1	0	
0	1	1	{	0	1	1	0	0	0	0	$F = 0$
				0	1	1	0	1	1	0	
1	0	0	{	1	0	0	0	0	0	0	$F = 0$
				1	0	0	0	1	1	0	
1	0	1	{	1	0	1	0	0	0	0	$F = D$
				1	0	1	0	1	1	1	
1	1	0	{	1	1	0	0	0	0	1	$F = 1$
				1	1	0	0	1	1	1	
1	1	1	{	1	1	1	0	0	0	1	$F = 1$
				1	1	1	0	1	1	1	

Step-4: Circuit Design



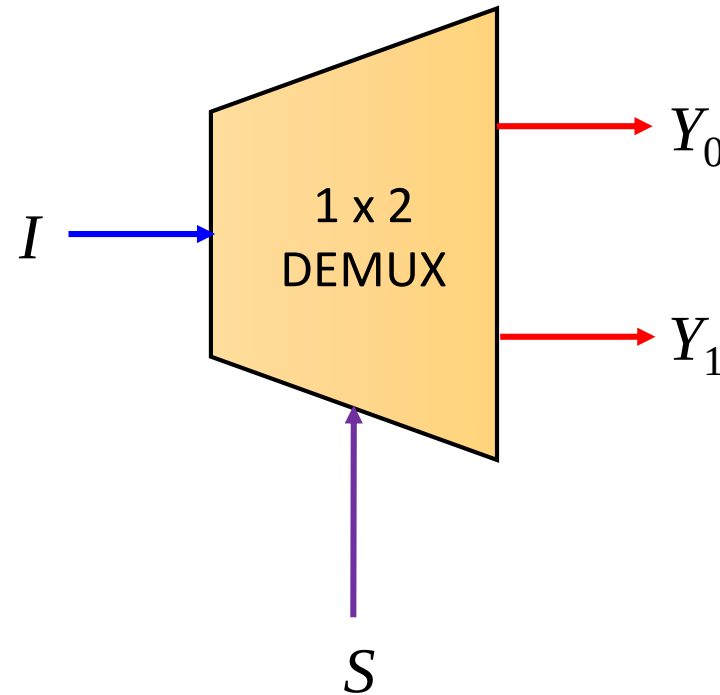
De-Multiplexer

DE-MULTIPLEXER

- ✓ A **demultiplexer** is a digital circuit that performs the **inverse operation of a multiplexer**.
- ✓ It has **1 input line and 2^n output lines** with **n selection inputs** to select the outputs.

1x2 Demultiplexer

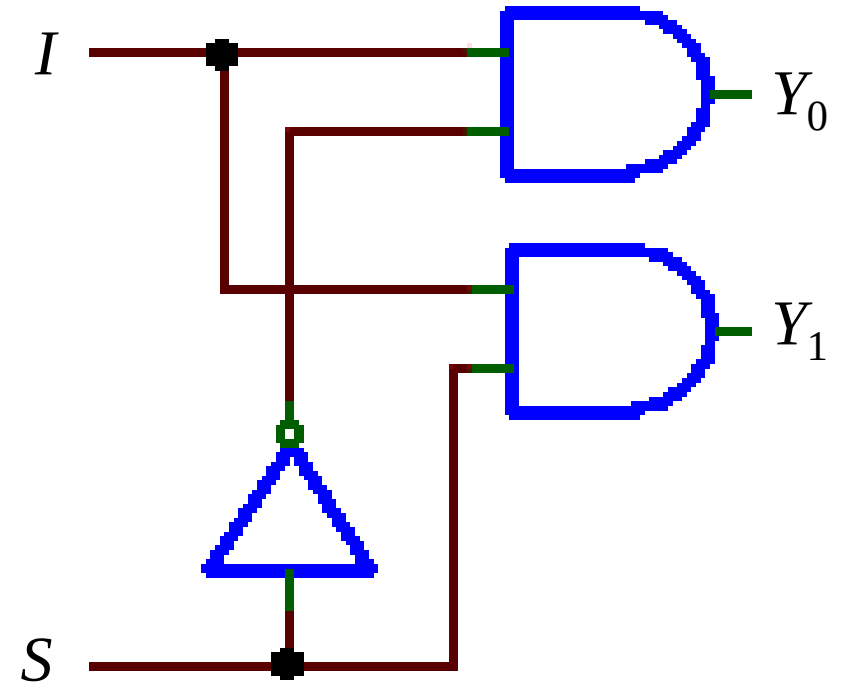
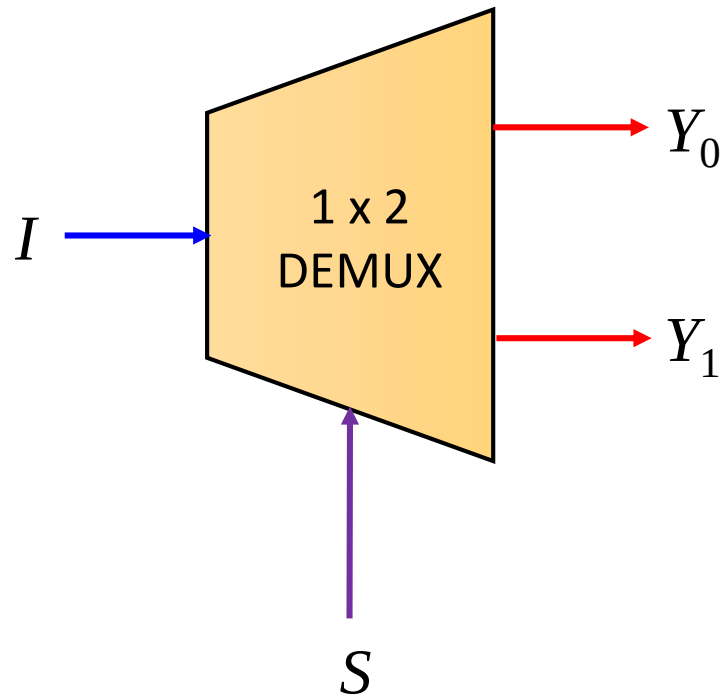
- ✓ It has **1** input line and **$2^1=2$** output Lines.
- ✓ It has **1** selection input **S** .



S	I	Y_0	Y_1
0	0	0	0
	1	1	0
1	0	0	0
	1	0	1

DE-MULTIPLEXER

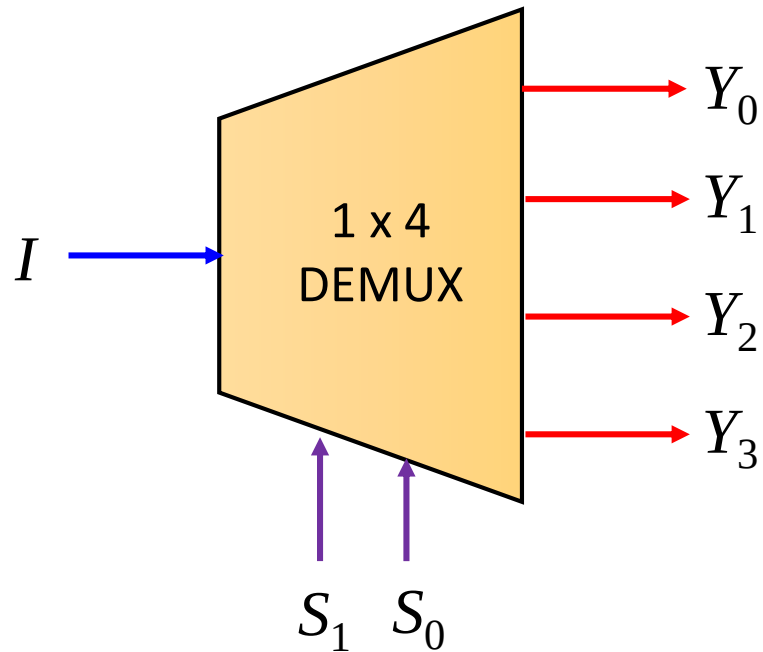
1x2 Demultiplexer



DE-MULTIPLEXER

1x4 Demultiplexer

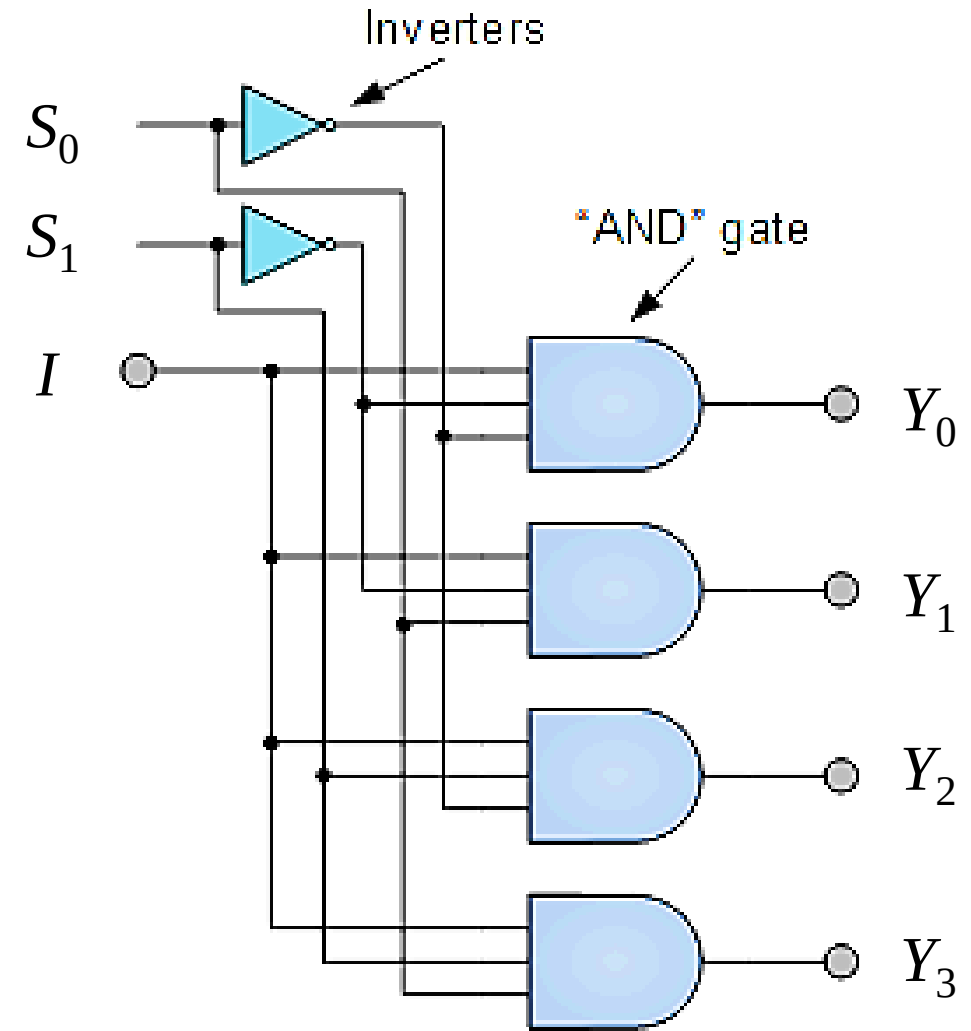
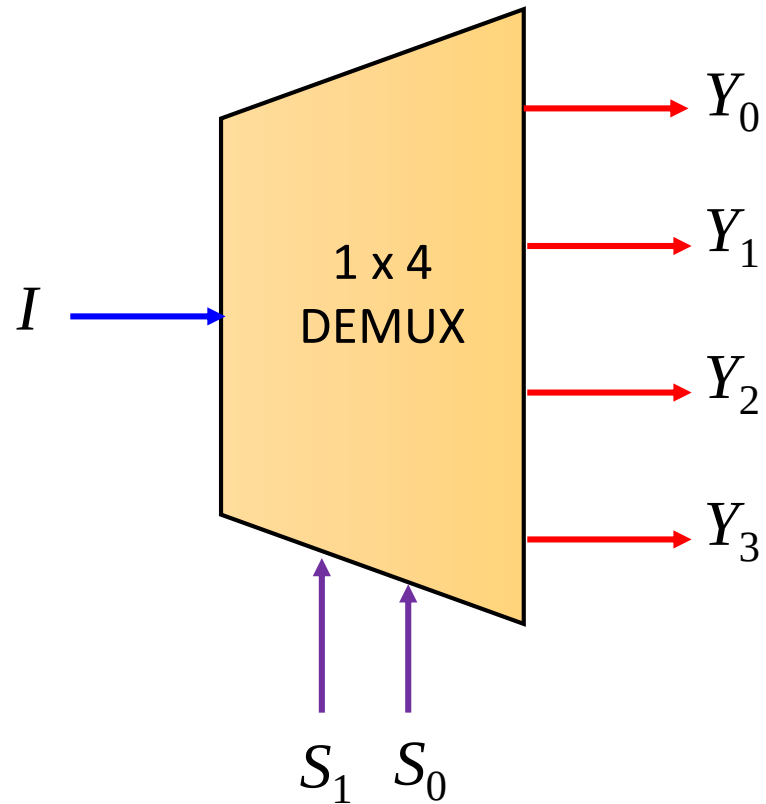
- ✓ It has **1** input line and **$2^2 = 4$** output Lines.
- ✓ It has **2** selection input **S_1** and **S_0** .



S_1	S_0	I	Y_0	Y_1	Y_2	Y_3
0	0	0	0	0	0	0
		1	1	0	0	0
0	1	0	0	0	0	0
		1	0	1	0	0
1	0	0	0	0	0	0
		1	0	0	1	0
1	1	0	0	0	0	0
		1	0	0	0	1

DE-MULTIPLEXER

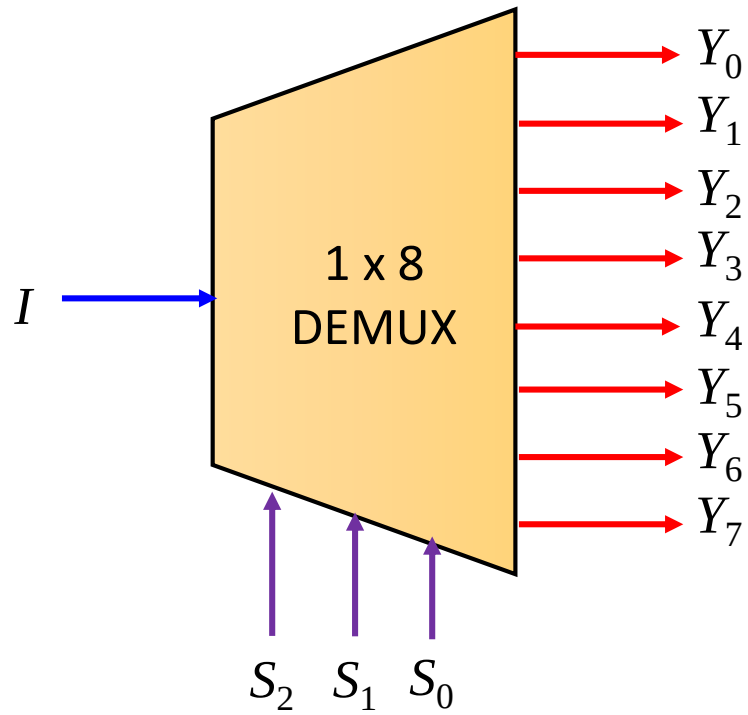
1x4 Demultiplexer



DE-MULTIPLEXER

1x8 Demultiplexer

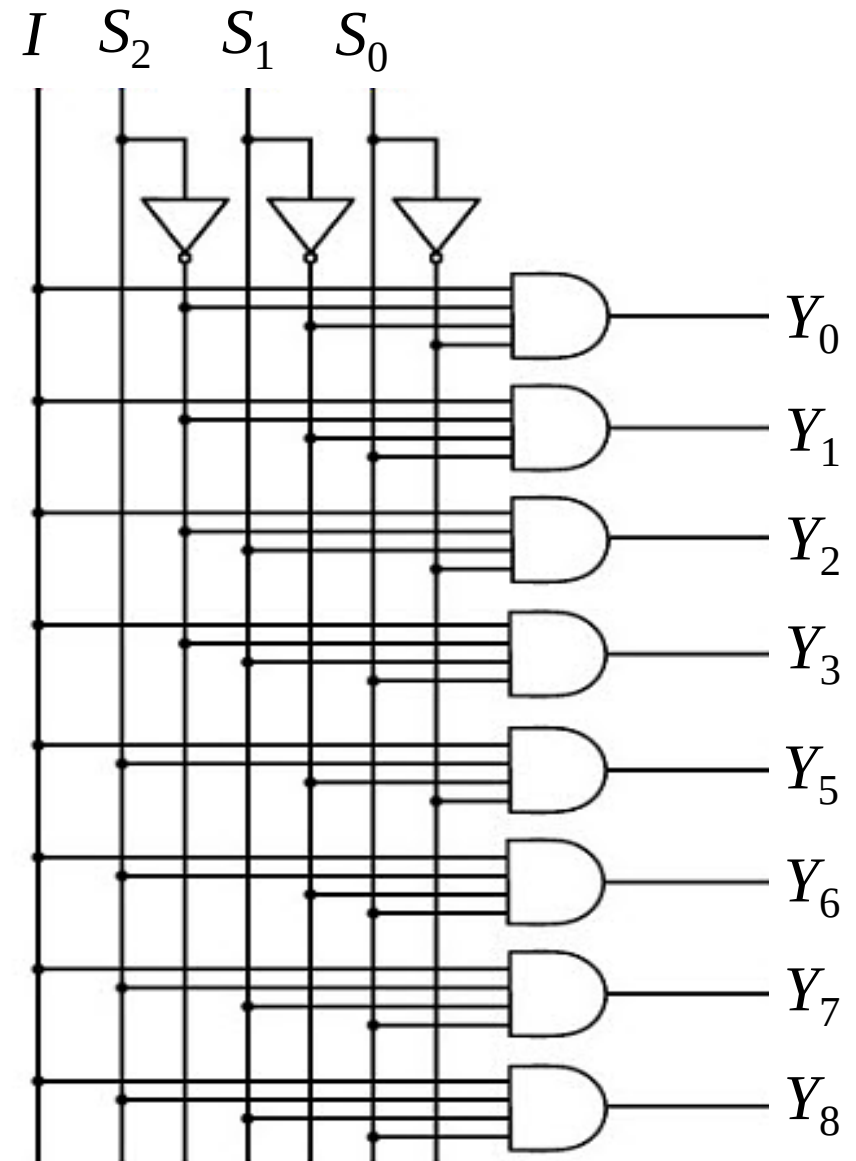
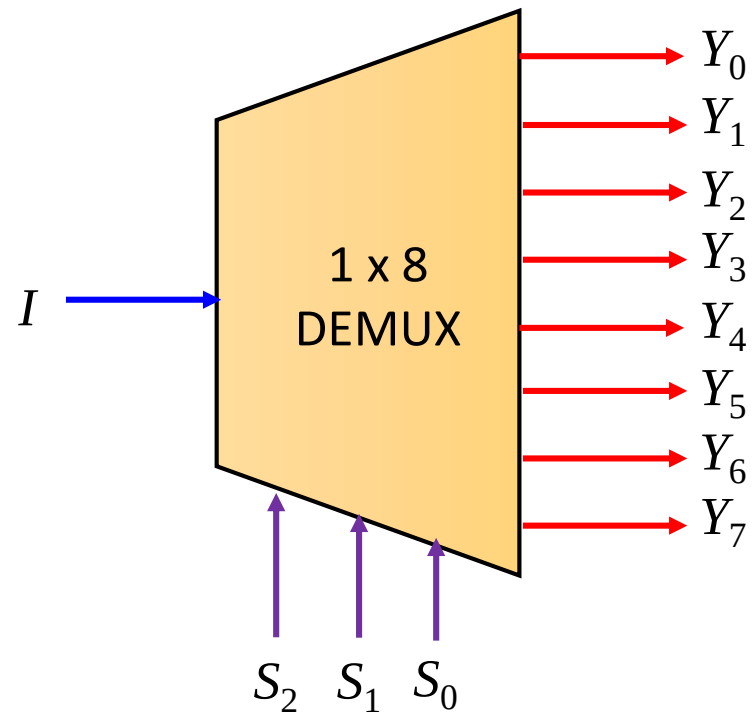
- ✓ It has **1** input line and **$2^3 = 8$** output Lines.
- ✓ It has **3** selection input S_2 , S_1 and S_0 .



S_2	S_1	S_0	I	Y_0	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	Y_7
0	0	0	0	0	0	0	0	0	0	0	0
			1	1	0	0	0	0	0	0	0
0	0	1	0	0	0	0	0	0	0	0	0
			1	0	1	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	0	0
			1	0	0	1	0	0	0	0	0
0	1	1	0	0	0	0	0	0	0	0	0
			1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	0
			1	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	0	0	0	0
			1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	0	0	0
			1	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	0	0
			1	0	0	0	0	0	0	0	1

DE-MULTIPLEXER

1x8 Demultiplexer



Thank You